9. Heavy Machinery Fleet & Site Equipment Operations

Document ID: DOC-09-MACH | **Version:** 3.7 (2026) | **Department:** Plant & Fleet Management
Primary OWL Classes: `ConstructionMachine`, `Excavator`, `Crane`, `Loader`, `Truck`, `Project`
Primary OWL Properties: `usesMachine`, `belongsToProject` (`owl:inverseOf usesMachine`)

9.1 Ontological Modeling of Construction Machinery

In `stavebna_firma_ontology - kópia.owl`, all motorized heavy equipment and site vehicles are represented under the class `ConstructionMachine`. The ontology enforces strong bidirectional semantics between projects and machinery:

Formal OWL Axioms for `ConstructionMachine` and Inverse Properties:

- **Universal Project Requirement:** Every `Project` (and specifically `BuildingProject`, `FamilyHouse`, `ApartmentBuilding`, `OfficeBuilding`, `IndustrialProject`, and `Reconstruction`) satisfies the restriction `usesMachine some ConstructionMachine`.
- **Universal Machine Assignment:** Every `ConstructionMachine` satisfies the restriction `belongsToProject some Project`.
- **Inverse Property Axiom (`owl:inverseOf`):** The object property `belongsToProject` (domain: `ConstructionMachine`, range: `Project`) is formally declared as the inverse of `usesMachine` (domain: `Project`, range: `ConstructionMachine`). Therefore, if `FamilyHouseExample usesMachine EXC_01`, an OWL reasoner automatically infers `EXC_01 belongsToProject FamilyHouseExample`!
- **Disjointness:** `ConstructionMachine` is disjoint with `ConstructionCompany`, `Person`, `Project`, `Material`, and `ConstructionPhase`.

9.2 The Four Subclasses of ConstructionMachine

The ontology classifies all heavy equipment into four specialized subclasses of `ConstructionMachine`:

1. **`Excavator` (Bager / Rýpadlo):** Tracked and wheeled hydraulic excavators used for earthmoving, trenching, foundation pit excavation, and demolition. Explicitly required by the ontology class `FamilyHouseExample` (`usesMachine some Excavator`).
2. **`Crane` (Žeriav):** Self-climbing tower cranes, saddle-jib tower cranes, and mobile telescopic all-terrain cranes used for vertical and horizontal lifting of `Steel`, formwork, precast `Concrete`, and `Glass` facade panels during `StructuralWork`.
3. **`Loader` (Nakladač):** Articulated front-end wheel loaders, skid-steer loaders, and backhoe loaders used for aggregate handling, site grading, backfilling, and pallet transport.
4. **`Truck` (Nákladné auto / Domiešavač):** Heavy 8x4 tipper dump trucks, flatbed material haulers, and transit concrete mixer trucks used to transport excavated soil and deliver bulk `Material`.

9.3 ApexBuild Active Machinery Fleet Register (2026)

Fleet ID	Ontology Subclass	Make & Model	Key Technical Specifications	Assigned Project (`belongsToProject`)	Internal Daily Rate

EXC-01	`Excavator`	Caterpillar 320 NextGen Crawler	Operating weight: 22.5 t | Bucket: 1.19 m³ | Max digging depth: 6.72 m | Engine: 121 kW Stage V	`FamilyHouseExample` (Záhorská Bystrica, 2026)	€680 / day
EXC-02	`Excavator`	Liebherr R 938 Litronic	Operating weight: 38.4 t | Heavy rock bucket: 2.2 m³ | Deep excavation configuration	`EuroveaTower` & Danube Atrium `OfficeBuilding`	€950 / day
CRN-01	`Crane`	Liebherr 280 EC-H 12 Litronic	Max lifting capacity: 12,000 kg | Jib radius: 70 m | Free-standing/ climbing height: up to 185 m	`EuroveaTower` (High-rise core & facade lifting)	€1,450 / day
CRN-02	`Crane`	Potain MDT 219 J10 Tower Crane	Max capacity: 10,000 kg | Jib length: 65 m | Anti-collision radar system installed	Danube Atrium `OfficeBuilding` & Trnava `IndustrialProject`	€1,100 / day
LDR-01	`Loader`	Volvo L120H Wheel Loader	Operating weight: 20.7 t | Bucket capacity: 3.8 m³ | Static tipping load: 14,500 kg	Trnava `IndustrialProject` & Hospital `Reconstruction`	€540 / day
TRK-01	`Truck`	Mercedes-Benz Arocs 4145 8x4 Tipper / Mixer	Gross vehicle weight: 41 t | 330 kW Euro 6d | 9 m³ Liebherr drum / 18 m³ heated steel dump body	Shared across `EuroveaTower`, `FamilyHouseExample` & `Reconstruction`	€490 / day

9.4 Machinery Maintenance & Wind-Speed Safety Thresholds

- **Tower Crane Wind Cut-Off:** All `Crane` units (CRN-01, CRN-02) are equipped with ultrasonic anemometers. Lifting of large-surface `Glass` panels or `Wood` CLT walls must cease when gusts exceed 13.8 m/s (50 km/h). At 20 m/s (72 km/h), all tower cranes must be placed in free-slewing weathervane mode under the supervision of the `SiteManager`.
- **Excavator Underground Utility Clearance:** Before any `Excavator` begins digging in the `Foundation` phase, the `SiteManager` must issue a signed Permit-to-Dig confirming ground-penetrating radar (GPR) clearance of gas, water, and high-voltage cables.